

Types of Computer Network

Q1. What is a computer network?

Ans: A computer network is a collection of interconnected computers, servers, mainframes, network devices, peripherals, or other devices connected to allow the sharing of data and resources.

Q2. What are the main types of computer networks based on geographical area?

Ans: The main types of computer networks based on geographical area are:
(i) Local Area Network (LAN) (ii) Metropolitan Area Network (MAN)
(iii) Wide Area Network (WAN)

Q3. What is a Local Area Network (LAN)?

Ans: A LAN is a network that connects devices within a limited area, such as a home, school, or single office building. It is characterized by high data transfer speeds and is typically privately owned.

Q4. What is a Metropolitan Area Network (MAN)?

Ans: A MAN is a network that covers a larger geographical area, such as an entire city or a large campus. It connects multiple LANs and is often managed by a consortium of users or a single network provider.

Q5. What is a Wide Area Network (WAN)?

Ans: A WAN covers a broad geographical area, such as a country, continent, or the entire globe. The Internet is the largest and most well-known example of a WAN.

Q6. What are the common network topologies used in computer networks?

Ans: Common network topologies include:
(i) Bus Topology (ii) Star Topology (iii) Ring Topology
(iv) Mesh Topology (v) Tree Topology

Q7. What is a Wireless Local Area Network (WLAN)?

Ans: A WLAN is a type of LAN that uses wireless technology, typically Wi-Fi, to connect devices to the network, allowing users to move around while staying connected.

Q8. What is a Storage Area Network (SAN)?

Ans: A SAN is a specialized, high-speed network that provides block-level access to storage, making the storage devices (like disk arrays) appear as locally attached to the servers.

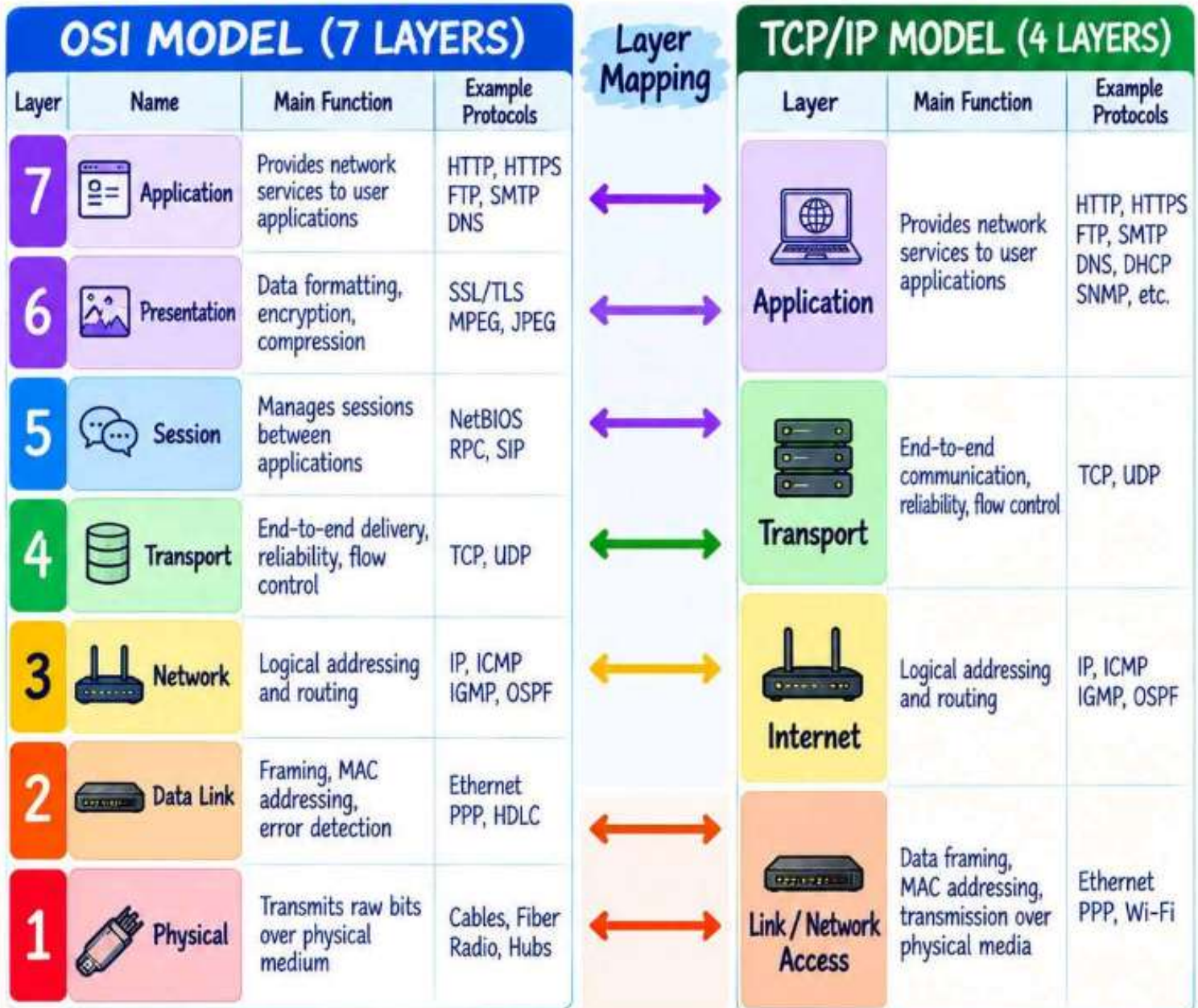
Q7. What is the difference between a Client-Server network and a Peer-to-Peer (P2P) network?

Q9. In a Client-Server network, specific computers (servers) provide resources and services to other computers (clients). In a P2P network, every computer acts as both a client and a server, sharing resources directly without a central server.

OSI Model *vs* TCP/IP Model

Connect
Learn
Grow

Different Models • Same Purpose
Understanding Networks Better



- 15.6 Best Practices for Network Security

16. Practical Network Setup

- 16.1 Home Network Setup: Step by Step
- 16.2 Small Office Network Setup
- 16.3 Basic Network Troubleshooting
- 16.4 Ping, Traceroute, and Other Tools

17. Network Commands and Tools

- 17.1 Windows Networking Commands
- 17.2 Linux/Unix Networking Commands
- 17.3 Basic Network Testing Tools
- 17.4 Capturing Network Traffic

18. Putting It All Together

- 18.1 How the Internet Works: End to End
- 18.2 Case Study: Loading a Web Page
- 18.3 Case Study: Sending an Email

19. Conclusion

- 11.3 Fiber Optic Cables
- 11.4 Wireless Transmission
- 11.5 Cable Categories and Speeds
- 11.6 Making and Testing Ethernet Cables

12. Client vs Server: Deep Dive

- 12.1 What is a Client? (Detailed)
- 12.2 What is a Server? (Detailed)
- 12.3 Types of Servers (Web, File, Email, Database, DNS, DHCP)
- 12.4 Server Hardware Considerations
- 12.5 The Client-Server Relationship (Detailed)

13. Peer-to-Peer vs Client-Server

- 13.1 Peer-to-Peer Networks: Detailed Analysis
- 13.2 Client-Server Networks: Detailed Analysis
- 13.3 Hybrid Approaches
- 13.4 Side-by-Side Comparison Table
- 13.5 Which Architecture to Choose?

14. Wireless Networking Fundamentals

- 14.1 How Wi-Fi Works
- 14.2 Wi-Fi Standards (802.11a/b/g/n/ac/ax)
- 14.3 Wi-Fi Frequency Bands (2.4 GHz vs 5 GHz vs 6 GHz)
- 14.4 Wi-Fi Security Protocols (WEP, WPA, WPA2, WPA3)
- 14.5 Setting Up a Wireless Network
- 14.6 Troubleshooting Wi-Fi Issues

15. Network Security Basics

- 15.1 Common Network Threats
- 15.2 Authentication Methods
- 15.3 Encryption: Symmetric vs Asymmetric
- 15.4 SSL/TLS: Securing Web Traffic
- 15.5 VPNs: Secure Tunnels

7. Ports and Protocols: The Rules of Communication

- 7.1 What is a Port?
- 7.2 Well-Known Ports (0-1023)
- 7.3 Registered Ports (1024-49151)
- 7.4 Dynamic/Private Ports (49152-65535)
- 7.5 Common Protocols and Their Ports
- 7.6 TCP vs UDP: The Two Transport Protocols

8. DHCP: Automatic IP Assignment

- 8.1 Why We Need DHCP
- 8.2 How DHCP Works: The DORA Process
- 8.3 DHCP Server Configuration
- 8.4 DHCP Relay Agents
- 8.5 DHCP Options and Scope

9. DNS: The Internet's Phonebook

- 9.1 Why DNS Exists
- 9.2 The DNS Hierarchy: Root, TLD, Authoritative
- 9.3 How DNS Resolution Works
- 9.4 DNS Record Types (A, AAAA, CNAME, MX, TXT)
- 9.5 DNS Caching and TTL
- 9.6 Configuring DNS on Different Systems

10. NAT: Network Address Translation

- 10.1 The IP Address Shortage Problem
- 10.2 How NAT Works: The Basics
- 10.3 Types of NAT: Static, Dynamic, PAT
- 10.4 NAT and Port Forwarding
- 10.5 NAT Traversal and Its Limitations

11. Network Media and Cabling

- 11.1 Twisted-Pair Cables (Ethernet)
- 11.2 Coaxial Cables

- 3.7 The Access Point: The Wireless Bridge
- 3.8 Modem: The Internet Gateway
- 3.9 Gateway: The Protocol Translator
- 3.10 Repeater: Signal Booster

4. The OSI Model: The Networking Blueprint

- 4.1 Why We Need a Model
- 4.2 Layer 1: Physical Layer
- 4.3 Layer 2: Data Link Layer
- 4.4 Layer 3: Network Layer
- 4.5 Layer 4: Transport Layer
- 4.6 Layer 5: Session Layer
- 4.7 Layer 6: Presentation Layer
- 4.8 Layer 7: Application Layer
- 4.9 How Data Moves Through the Layers (Encapsulation)

5. The TCP/IP Model: The Internet's Foundation

- 5.1 OSI vs TCP/IP: Understanding the Difference
- 5.2 Layer 1: Network Interface Layer
- 5.3 Layer 2: Internet Layer
- 5.4 Layer 3: Transport Layer
- 5.5 Layer 4: Application Layer
- 5.6 The TCP/IP Protocol Suite

6. IP Addressing: The Digital Postal System

- 6.1 What is an IP Address?
- 6.2 IPv4 Addresses: Structure and Classes
- 6.3 Public vs Private IP Addresses
- 6.4 Subnetting: Dividing Networks
- 6.5 Subnet Masks Explained
- 6.6 CIDR Notation: Modern Addressing
- 6.7 IPv6: The Next Generation
- 6.8 IPv4 vs IPv6: Key Differences

Networking Fundamentals

A Complete Beginner's Journey

Table of Contents

1. What is a Network?

- 1.1 Defining a Network in Simple Terms
- 1.2 The History of Networking: From Telegraph to Internet
- 1.3 Why Networks Exist: The Core Purpose
- 1.4 Key Components of Any Network
- 1.5 How Data Travels: The Journey of a Message
- 1.6 Bandwidth, Throughput, and Latency Explained
- 1.7 Network Topologies: How Devices Are Arranged

2. Types of Networks (LAN, WAN, MAN, PAN, CAN, SAN)

- 2.1 Personal Area Network (PAN)
- 2.2 Local Area Network (LAN)
- 2.3 Campus Area Network (CAN)
- 2.4 Metropolitan Area Network (MAN)
- 2.5 Wide Area Network (WAN)
- 2.6 Storage Area Network (SAN)
- 2.7 Virtual Private Network (VPN)
- 2.8 Comparison Summary Table

3. Network Devices - Deep Dive

- 3.1 Network Interface Card (NIC)
 - 3.2 The Hub: The Simplest Device
 - 3.3 The Bridge: Connecting Network Segments
 - 3.4 The Switch: The Smarter Hub
 - 3.5 The Router: The Traffic Director
 - 3.6 The Firewall: The Security Guard
-

HOW TO CONFIGURE OSPF ON A CISCO ROUTER

SINGLE AREA OSPF (AREA 0) – STEP-BY-STEP GUIDE



WHAT IS OSPF?



OSPF (Open Shortest Path First) is a dynamic routing protocol that automatically learns and shares routes between routers.

BENEFITS

- ✓ Fast Convergence
- ✓ Scalable
- ✓ Open Standard
- ✓ Supports VLSM & CIDR

NETWORK TOPOLOGY



1



ENTER PRIVILEGED MODE

```
Router> enable
```

Access privileged EXEC mode.

2



ENTER CONFIGURATION MODE

```
Router# configure terminal
```

Enter global configuration mode.

3



START OSPF PROCESS

```
Router(config)# router ospf 1
```

Start OSPF process.
1 = Process ID

4



ADVERTISE NETWORK 1 (LAN 1)

```
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
```

Advertise LAN 1 network.

5



ADVERTISE WAN LINK

```
Router(config-router)# network 10.10.10.0 0.0.0.3 area 0
```

Advertise WAN link network.

6



ADVERTISE NETWORK 2 (LAN 2)

```
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
```

Advertise LAN 2 network.

7



CONFIGURE ROUTER ID (OPTIONAL)

```
Router(config-router)# router-id 1.1.1.1
```

Set a unique Router ID.

8



SAVE CONFIGURATION

```
Router# copy running-config startup-config
```

Save running configuration to NVRAM.

COMPLETE CONFIGURATION EXAMPLE

ROUTER R1

```
enable
configure terminal

router ospf 1
router-id 1.1.1.1

network 192.168.1.0 0.0.0.255 area 0
network 10.10.10.0 0.0.0.3 area 0

end
copy running-config startup-config
```

R1

ROUTER R2

```
enable
configure terminal

router ospf 1
router-id 2.2.2.2

network 192.168.2.0 0.0.0.255 area 0
network 10.10.10.0 0.0.0.3 area 0

end
copy running-config startup-config
```

R2

VERIFY OSPF

- | | | |
|-----|------------------------------------|-----------------------------|
| >.. | <code>show ip ospf neighbor</code> | Check OSPF neighbor status. |
| >.. | <code>show ip route ospf</code> | View OSPF learned routes. |
| >.. | <code>show ip protocols</code> | Verify OSPF configuration. |

OSPF NEIGHBOR STATES



When neighbors reach FULL state, routing information is fully exchanged.

KEY BENEFITS OF OSPF

- | | | | | |
|--------------------------|------------------|-------------------------|-------------------|----------------------------|
| Automatic Route Learning | Fast Convergence | Supports Large Networks | Loop-Free Routing | Industry Standard Protocol |
|--------------------------|------------------|-------------------------|-------------------|----------------------------|



OSPF helps routers find the best path automatically and keeps your network fast, efficient & reliable.

≡ NETWORK RACK REPLICA ≡



SWITCH

Connects devices within the network and manages data traffic.



FABRIC INTERCONNECT

High-speed connections between network devices.



NAS NAS SWITCH

Centralized storage and file sharing over the network.



PDU

Distributes power to all devices in the rack safely.



UPS

Provides backup power and protects against outages.



PATCH PANELS

Organizes and manages cables for easy connectivity.



FIREWALL

Monitors and controls incoming and outgoing network traffic.



SERVERS

Hosts applications, services and data for the network.



KVM

Allows remote management of servers and devices.



www.thenetworkdna.com



menjadi jembatan server remote (SSH tunnel)
 Sedangkan :

~ ponsel Vivo indikator (Streamlit / UI)
 penunjang koneksi ke streamlit dengan
 tampilan layar hacker di TV 40 Inch

Peran fundamental 2 ponsel :

- Vivo Y28 (64 bit) AI inference

2> AI lokal
 diinstall
 1 atau 3 ?
 Emulasi

* penggerak otak AI
 lokal Jarvis Mico
 V6.0

- Infinix [32 bit] Log Server / Gateway
 < Eksekutor > ~ Simpan catatan (Log)
 dan firewall jalur sederhana
 < Assisten >

catatan tambahan USB flash disc

1. flash disc A GB : ADB Tools &
 Driver

~ Komunikasi lancar Vivo & Infinix

2. flash disc B GB : Portable Python
 (win python) Main.py kontrol HP
 dari Laptop tanpa perlu install HP

1) ADB Remote Controller & file Bridge

~ Laptop Travelmate menyalakan 2 ponsel dengan ADB (Android Debug Bridge)

* fungsi Laptop :

~ Monitor 2 ponsel memindahkan file dari laptop ke termux [vivo + infinix] tanpa wifi [Power shell masing-masing]

* Reverse Tethering : jika ada wifi sumber internet bisa jadi jalur online untuk 2 ponsel mendapatkan sumber internet [online meskipun tanpa internet]

Pertanyaan :

1. jika free wifi ?

2. jika tethering atau mode transfer file

3. jika laptop mati solusi ?

* Distributed "Kabel Ghoib" Hub

~ Tailscale atau VPN Tunneling :

* Ponsel memiliki modem internal jadi Gateway

1) Ponsel Infinix bertugas

Hari

Tanggal

Tahun

KTP, NPWP, Ijazah, foto

Cara kerja menemukan ponsel Apple

1. Harus perangkat Apple yang lewat (jabat)
2. Harus menyala minimal 10 detik
3. Walaupun kartu SIM GSM masih bisa dihack dari IMEI

A. jika dimasukkan kartu baru bisa deteksi dari IP Address yg baru

Langkah - Langkah
Hard Ware

- Dual Mode Processing Master & Node
1. Stabilitas : stabil power listrik
 2. Kabel USB Port hub + flash disc
= Ready Boost Windows [swap windows]
 3. Peran 3 fungsi strategis 2 ponsel :
 - a. Vivo Y28 [64 Bit : The Heavy Lifter] Model AI modern / printer Agent & memproses logika sensor Software Engineer [Binder Micro komputer]
 - b. Infinix [32 bit] the Legacy / IOT controller] pengatur protokol IOT [Masquito MQTT] sensor tugas sepele